

AUTOMATED SPEECH RECOGNITION

Model	Algorithms Used	Advantages	Disadvantages	Limitations	Results / Performance Examples
HMM-Based Model	Hidden Markov Model (HMM) combined with Gaussian Mixed Model (GMM) or Deep Neural Network (DNN); Bayesian theory; acoustic model $P(X S)$, pronunciation model $p(S L)$, and language model $p(L)$	Historically mainstream; state-of-the-art performance with deep learning; uses dictionaries and large external language models for wide linguistic coverage	Complex training process with multiple modules; individual optimality doesn't guarantee global optimality; relies on conditional independence assumptions	Bottleneck reached; performance comparable to HMM-DNN; internal representation requires transcription processing	Long-standing industry standard; outperformed by some E2E models in specific contexts but still strong
End-to-End Model (General)	Directly maps audio to words/graphemes using one deep network; encoder, aligner, and decoder	Simplified model and joint training; global optimization possible; direct transcription without forced alignment; no domain expertise required	Performance often worse than or comparable to HMM-DNN; language knowledge limited to training transcriptions	Current performance bottlenecked by training data transcriptions	Simplified architecture but needs more data and language integration
CTC-Based End-to-End Model	Connectionist Temporal Classification (CTC) loss; 'blank' labels and path aggregation; neural network acoustic models (CNN/RNN)	Eliminates data segmentation alignment; shortest delay among E2E models; low computational complexity $O(T)$	Assumes output labels are independent (no modeling of interdependencies); cannot inherently learn language knowledge; worst performance without external LM	Cannot map to sequences longer than input; needs external LM for accuracy	WSJ dataset: 30.1% WER alone, 8.7% with trigram LM; CTC-Grapheme WER: 39.4%
RNN-Transducer End-to-End Model	Transcription, Prediction (RNN), and Joint networks; 'blank' labels and path aggregation	Maps to arbitrary lengths; prediction network learns language interdependencies; supports streaming ASR	Difficult to train (needs pre-training); higher delay and complexity $O(TN)$ than CTC	Language knowledge coverage limited to transcriptions	TIMIT: 17.7% PER; Google Voice Search: 8.5% WER; RNN Transducer WER: 6.6%